MARKING SCHEME

Model Paper — 02

Mathematics II

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Notes (for teachers):

- Paper II: candidates answer **five** questions from Part A (Questions 1 – 6) and **five** questions from Part B (Questions 7 – 12). Each question carries 10 marks; total 100.
- The marks for each step are given in the right-hand column and add up to the marks printed against each sub-part of the question paper.
- Give full marks for correct alternative methods. Accept reasonable readings from graphs and measured values from constructions (acceptance ranges are shown). Where tables are used, 4-figure table values are expected.

Part A — Answers and mark allocation

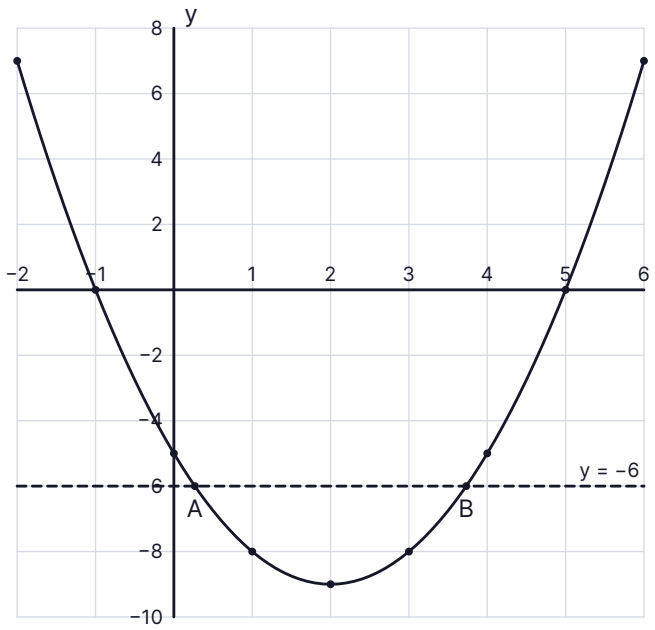
Question 1

(a)	
Interest = $\frac{14}{100} \times 60\,000$ = Rs. 8 400	**1**
Amount to settle = 60 000 + 8 400 = Rs. 68 400	**1**
(b)	
Total paid = 60 000 + 5 460 = Rs. 65 460	**1**
One instalment = 65 460 ÷ 12 = Rs. 5 455	**1**
(c)	
Unpaid balance in the 12 months: 60 000, 55 000, 50 000, … , 10 000, 5 000	**1**
Sum of the balances = 5 000 × (12 + 11 + … + 1) = 5 000 × 78 = Rs. 390 000 (rupee-months)	**1**
Converted to years: 390 000 ÷ 12 = Rs. 32 500 for one year (average balance)	**1**
Annual rate = $\frac{5\,460}{32\,500} \times 100$	**1**
= 16.8%	**1**
Amaya's rate is 14% and 16.8% > 14%, so Pradeep's rate is higher	**1**

Total: 10

Question 2

(i)	
Axis of symmetry is $x = 2$, so the value at $x = 4$ equals the value at $x = 0$: $y = -5$	**1**
(ii)	
Axes drawn with a suitable scale (e.g. 2 cm for 1 unit on x, 1 cm for 1 unit on y)	**1**
All points (−2, 7), (−1, 0), (0, −5), (1, −8), (2, −9), (3, −8), (4, −5), (5, 0), (6, 7) plotted correctly	**1**
Joined by a smooth U-shaped curve with minimum point (2, −9)	**1**
(iii)	
(a) $x = -1$ and $x = 5$	**1**
(b) $y = (x + 1)(x - 5)$	**1**
(iv)	
(a) Line $y = -6$ drawn; A at $x \approx 0.3$ and B at $x \approx 3.7$ read from the graph (accept 0.2 – 0.4 and 3.6 – 3.8)	**1**
Range: $0.3 < x < 3.7$	**1**
(b) AB = 3.7 − 0.3 (readings from the graph)	**1**
AB ≈ 3.5 units (accept 3.4 – 3.6; exact value $2\sqrt{3} = 3.46$)	**1**

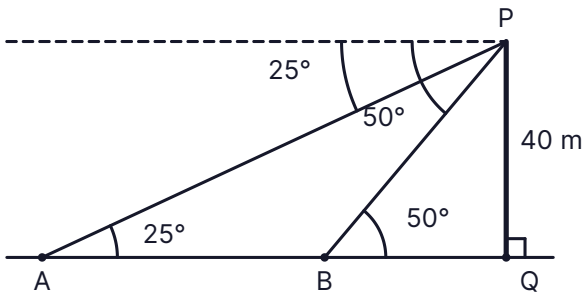


Completed graph: minimum point (2, −9); cuts the x-axis at −1 and 5; the line $y = -6$ meets it at A and B

Total: 10

Question 3

(i)	
Horizontal through P drawn; angles of depression marked at P (to A and to B)	**1**
PÂQ = 25°, PB̂Q = 50° (alternate angles, horizontal through P ∥ AQ)	**1**
(ii)	
$\tan 25° = \frac{40}{AQ}$	**1**
AQ = 40 ÷ 0.4663 = 85.78 m	**1**
$\tan 50° = \frac{40}{BQ}$	**1**
BQ = 40 ÷ 1.1918 = 33.56 m	**1**
(iii)	
A, B and Q are in a line with B between A and Q, so AB = AQ − BQ	**1**
AB = 85.78 − 33.56	**1**
= 52.22 m	**1**
52.22 > 52, so the distance is greater than 52 m	**1**



Angles of depression 25° and 50° at P; equal angles 25° and 50° at A and B

Total: 10

Question 4

(i)	
Area of rectangle ABCD = $(2x + 3)(x + 6) = 2x^2 + 15x + 18$	**1**
Area of metal = $2x^2 + 15x + 18 - x^2 = x^2 + 15x + 18$	**1**
$x^2 + 15x + 18 = 93 \Rightarrow x^2 + 15x - 75 = 0$	**1**
(ii)	
$x = \frac{-15 \pm \sqrt{(225 + 300)}}{2} = \frac{-15 \pm \sqrt{525}}{2}$	**1**
$\sqrt{525} = 5\sqrt{21} = 5 \times 4.58 = 22.9$, so $x = \frac{-15 + 22.9}{2} = 3.95$ (the negative root is rejected)	**1**
$x \approx 4$ cm	**1**
(iii)	
Perimeter $= 2[(2x + 3) + (x + 6)] = 6x + 18$	**1**
$x = \frac{-15 + 5\sqrt{21}}{2}$, so $6x + 18 = 3(-15 + 5\sqrt{21}) + 18$	**1**
$= 15\sqrt{21} - 27$ cm	**1**
$= 15 \times 4.58 - 27 = 41.7 > 40$	**1**

Total: 10

Question 5

(i)	
$x + y = 40$	**1**
$90x + 60y = 2\,850$, i.e. $3x + 2y = 95$	**1**
Substituting $y = 40 - x$: $3x + 80 - 2x = 95$	**1**
$x = 15$ adults	**1**
$y = 25$ children	**1**
(ii)	
Let each adult carry n books; each child carries $(n - 2)$ books: $15n + 25(n - 2) = 190$	**2**
$15n + 25n - 50 = 190$	**1**
$40n = 240$	**1**
$n = 6$ books	**1**

Total: 10

Question 6

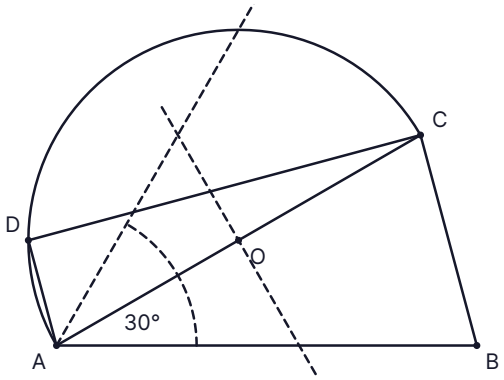
(i)	
Mid-values 14.5, 24.5, 34.5, 44.5, 54.5, 64.5	**1**
Σfx = 14.5 + 49 + 207 + 356 + 381.5 + 387 = 1 395 (Σf = 30)	**1**
Mean = 1 395 ÷ 30 = 46.5 books	**1**
(ii)	
Expected number = 46.5 × 24	**1**
= 1 116 books	**1**
(iii)	
Expected income = 1 116 × Rs. 5	**2**
= Rs. 5 580	**1**
(iv)	
Days with fewer than 30 books: classes 10 – 19 (1 day) and 20 – 29 (2 days); minimum total = 1 × 10 + 2 × 20	**1**
= 50 books	**1**

Total: 10

Part B — Answers and mark allocation

Question 7

(i)	
AB = 9 cm drawn	**1**
60° at A constructed (equilateral-triangle arcs, arc marks visible)	**1**
The 60° angle bisected (angle of 30° with AB)	**1**
(ii)	
C marked on the bisector with AC = 9 cm (compass arc from A)	**1**
BC joined (triangle ABC isosceles, AB = AC, apex angle 30°; ∠ABC = ∠ACB = 75°)	**1**
(iii)	
Perpendicular bisector of AC with arcs visible; mid-point O of AC	**1**
Semicircle of radius 4.5 cm with centre O on the side of AC opposite to B	**1**
(iv)	
Line through A parallel to BC, constructed by copying ∠ACB (or ∠ABC) with arcs visible	**1**
D marked where it meets the semicircle, CD joined (AD ≈ 2.3 cm, CD ≈ 8.7 cm; accept ±0.2 cm)	**1**
(v)	
$A\hat{D}C = 90°$ (angle in a semicircle); $D\hat{A}C = A\hat{C}B = 75°$ (alternate angles, AD ∥ BC); so $A\hat{C}D = 180° − 90° − 75° = 15°$ (accept a measured value 14° – 16°)	**1**



Completed construction (the dashed lines are construction lines)

Total: 10

Question 8

(i)	
In $\triangle BGE$ and $\triangle CFE$: BE = EC (E is the mid-point of BC)	**1**
$B\hat{E}G = C\hat{E}F$ (vertically opposite angles)	**1**
$E\hat{B}G = E\hat{C}F$ (alternate angles, AG ∥ DC); so $\triangle BGE \equiv \triangle CFE$ (angle, side, angle)	**1**
(ii)	
GE = EF (corresponding sides of congruent triangles) and BE = EC	**1**
The diagonals BC and GF of BGCF bisect each other at E, so BGCF is a parallelogram	**1**
(iii)	
In $\triangle DFH$ and $\triangle BGE$: $D\hat{F}H = B\hat{G}E$ (corresponding angles, DC ∥ AG)	**1**
$D\hat{H}F = B\hat{E}G$ (corresponding angles, AH ∥ BC); the third angles are equal, so the triangles are equiangular	**1**
(iv)	
$\frac{DH}{BE} = \frac{DF}{BG}$ (corresponding sides of equiangular triangles)	**1**
BG = CF (congruent triangles) = $\frac{3}{4}DC = \frac{3}{4}AB$, and DF = $\frac{1}{4}AB$; so $\frac{DF}{BG} = \frac{1}{3}$	**1**
BE = $\frac{1}{2}BC = \frac{1}{2}AD$, so DH = $\frac{1}{3} \times \frac{1}{2}AD = \frac{1}{6}AD$	**1**

Total: 10

Question 9

(i)	
Volume of the cylinder = $\pi r^2 \times 2r = 2\pi r^3$	**1**
Volume of one pyramid = $\frac{1}{3} \times a^2 \times 3a = a^3$	**1**
Volume of 6 pyramids = $6a^3$	**1**
$6a^3 = 2\pi r^3 \Rightarrow a^3 = \frac{\pi}{3} r^3$	**1**
(ii)	
$a^3 = \frac{3.14 \times 8.1^3}{3}$; lg a^3 = lg 3.14 + 3 lg 8.1 − lg 3	**1**
= 0.4969 + 3 × 0.9085 − 0.4771 = 2.7453	**1**
a^3 = 556.3 (accept 556.2 – 556.3); lg a = 0.9151 ⇒ a = 8.22 ≈ 8 cm	**1**
(iii)	
Volume of the cuboid = $a^2 \times 3a = 3a^3 = 3 \times 8^3$ = 1 536 cm^3	**1**
Volume of the pyramid = a^3 = 512 cm^3	**1**
Remaining metal = 1 536 − 512 = 1 024 cm^3	**1**

Total: 10

Question 10

(i)	
15, 18, 21	**1**
(ii)	
$T_{16} = a + 15d = 15 + 15 \times 3$	**1**
= 60 books	**1**
(iii)	
$S_{16} = \frac{16}{2}(15 + 60)$	**1**
= 600 books	**1**
(iv)	
$\frac{15}{2}[2a + 14 \times 2] = 600$	**1**
$2a + 28 = 80$	**1**
a = 26 books	**1**
(v)	
$15 + 3(n - 1) = 26 + 2(n - 1)$, i.e. $3n + 12 = 2n + 24$	**1**
n = 12, the 12th day (each arranges 48 books)	**1**

Total: 10

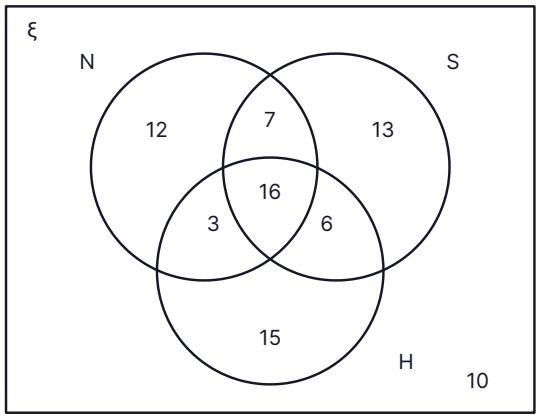
Question 11

(i)(a)	
$O\hat{B}T = 90°$ and $O\hat{C}T = 90°$ (a tangent is perpendicular to the radius at the point of contact)	**1**
$O\hat{B}T + O\hat{C}T = 180°$, a pair of opposite angles supplementary, so OBTC is a cyclic quadrilateral	**1**
(i)(b)	
OB = OC (radii) and TB = TC (tangents from T), so OT is the perpendicular bisector of BC; hence OT $\perp$ BC (or $\triangle OBT \equiv \triangle OCT$ and $B\hat{O}T = C\hat{O}T$)	**2**
$A\hat{C}B = 90°$ (angle in a semicircle), so AC $\perp$ BC	**1**
OT and AC are both perpendicular to BC (corresponding angles equal), so OT $\parallel$ AC	**1**
(ii)	
$T\hat{B}C = T\hat{C}B$, since TB = TC (tangents from T; isosceles triangle)	**2**
$T\hat{B}C = B\hat{A}C$ (angle in the alternate segment). (Also accepted: $B\hat{O}T$, with the reason that $B\hat{O}T = \frac{1}{2} B\hat{O}C = B\hat{A}C$)	**2**

Total: 10

Question 12

(i)	
Members who borrowed at least one type = 82 − 10 = 72	**1**
Exactly one type = 72 − 32 (32 borrowed two or more types)	**1**
= 40	**1**
(ii)	
Let the number who borrowed all three types be g; exactly two types = g	**1**
$g + g = 32$	**1**
$g = 16$	**1**
(iii)	
Exactly two types = 16, of which Novels & Science only = 7; let Novels & History only = y, Science & History only = $2y$: $7 + 3y = 16$, $y = 3$ (Novels & History only = 3, Science & History only = 6)	**1**
History only = 40 − (6 + 3 + 16) = 15	**1**
(iv)	
Novels only = 38 − (7 + 3 + 16) = 12	**1**
Science only = 40 − 12 − 15 = 13	**1**



Completed Venn diagram (Novels only 12, Science only 13, History only 15, N∩S only 7, S∩H only 6, N∩H only 3, all three 16, none 10)

Total: 10